Pathway and Network Analysis of UPF3B Knockout

in Neural Progenitor Cells

BCB420 project presentation

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Introduction - *background*

1. Biological Background (Tan et al. 2025)

- Nonsense-mediated mRNA decay (NMD) (Karousis et al. 2016)
- Regulates mRNA stability and overall gene expression
- NMD sensitivity depends on cellular context, not only transcript features

2. Role of UPF3B in the Nervous System

UPF3B plays critical roles in nervous system function (Kadlec et al. 2004)

- Mutations associated with: Intellectual disability; Schizophrenia, ASD, ADHD
- Loss of UPF3B:

i) Impairs neural differentiation

ii) Affects neuronal maturation

- Proposed mechanism:

i) Regulates a specific subset of NMD-target mRNAs

ii) Essential for normal neural development

Introduction - *dataset selection*

3. Dataset Overview (GSE263406) (Tan et al. 2025)

- Data type: RNA-seq (Illumina NovaSeq 6000)
- Total: 43,884 genes × 12 samples
- in Neural progenitor cells (NPCs)
- **Ctrl** control (n=6) vs **UPF3B-KO** knockout (n=6)

4. Research Goal

- Identify biological pathways affected by UPF3B in NPCs
- Focus on neurodevelopment-related pathways
- Use pathway analyses (ORA/GSEA) to connect mRNA changes to functional outcomes

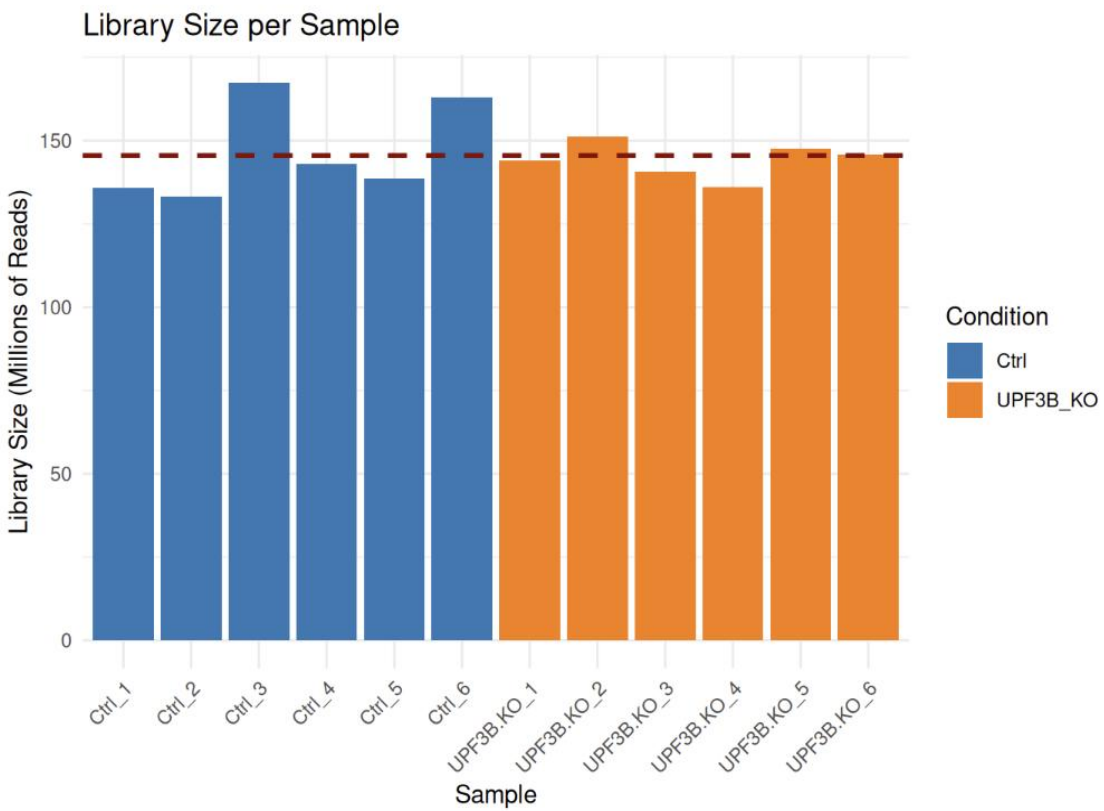
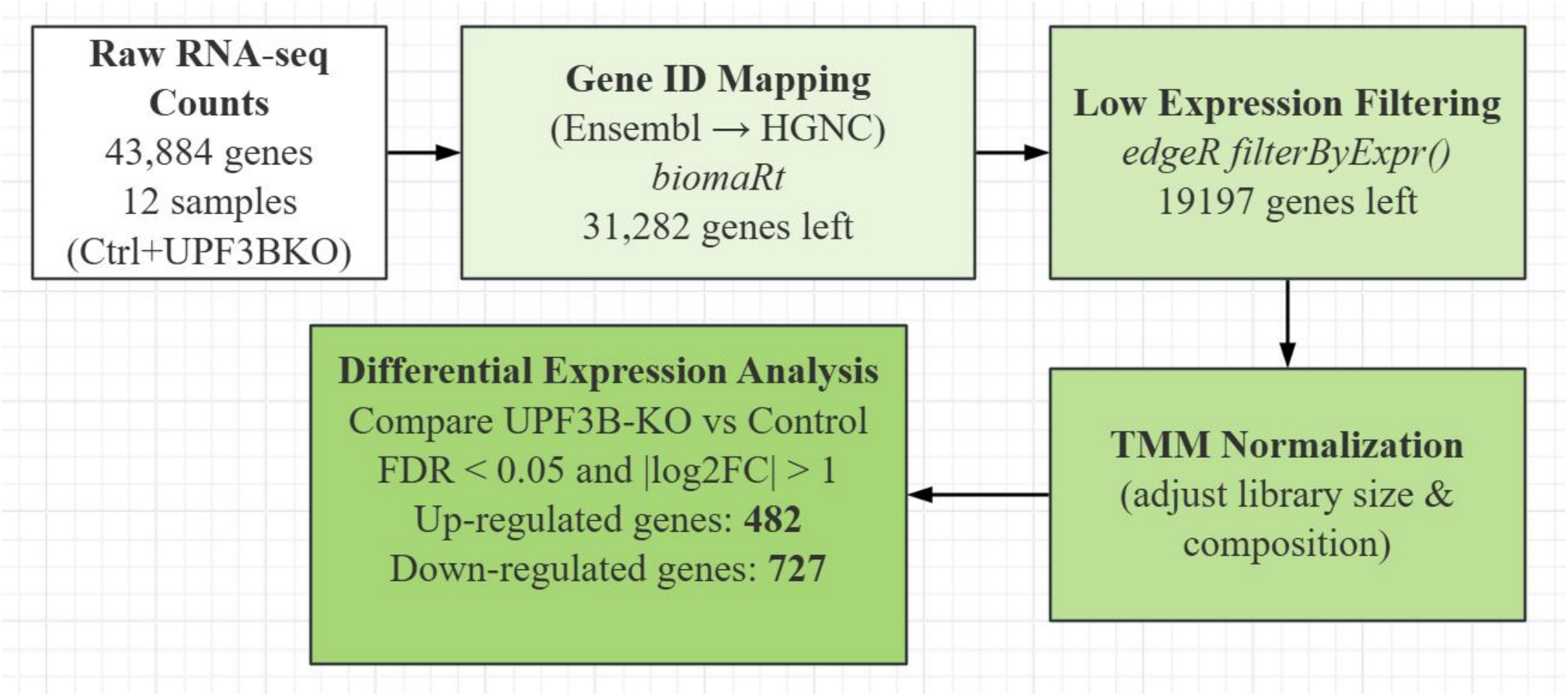


Figure 1. Library sizes of each RNA-seq sample in the GSE263406 dataset. Total read counts for each sample are shown in millions, with colors indicating experimental condition (blue = Control, orange = UPF3B knockout).

Methods - *Pre-Pathway Analysis Workflow*



Methods - *Pre-Pathway Analysis Workflow*

Differential Expression Analysis
Compare UPF3B-KO vs Control
FDR < 0.05 and |log2FC| > 1
Up-regulated genes: **482**
Down-regulated genes: **727**

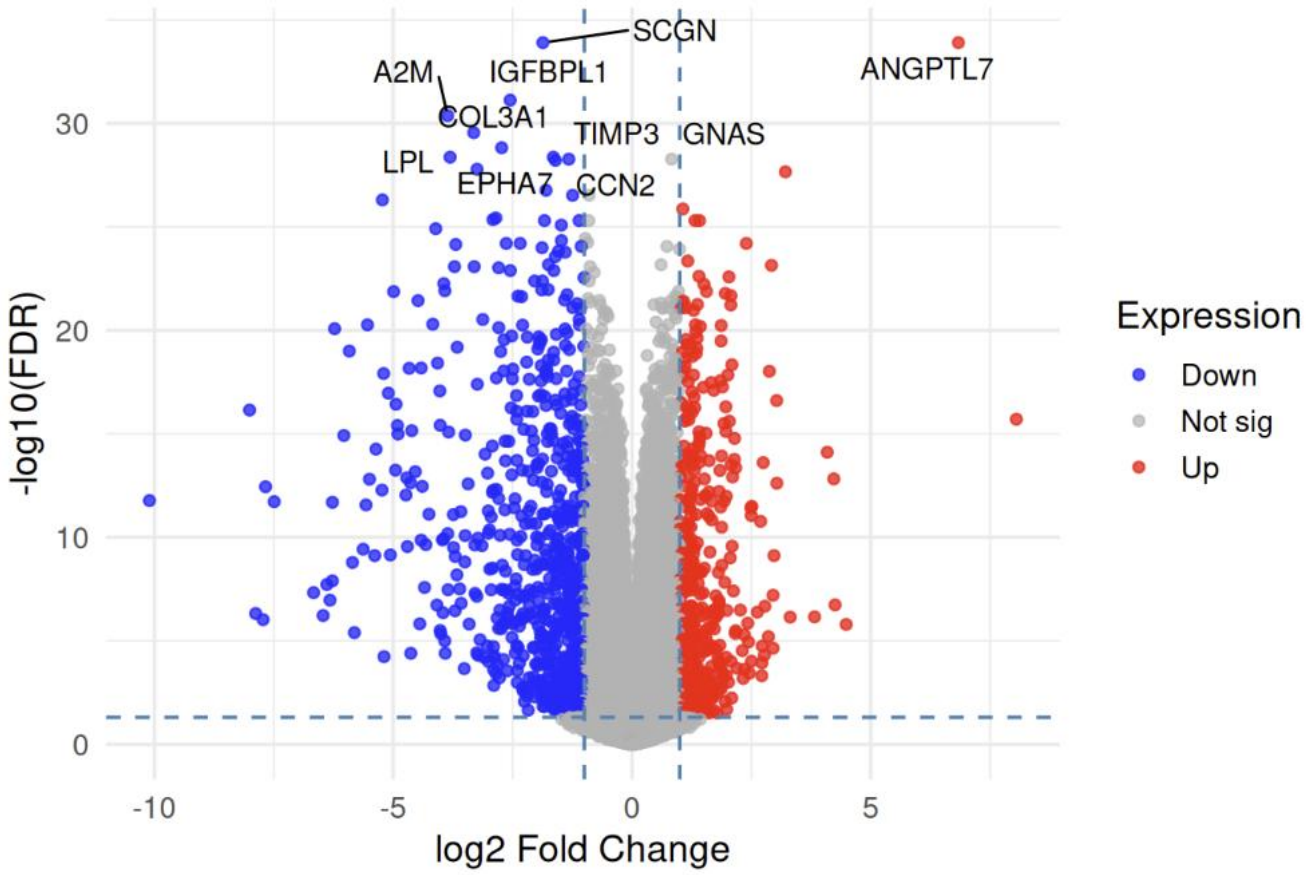


Figure 2. Volcano plot of differential gene expression between UPF3B_KO and Ctrl samples. Each point represents a gene, with the x-axis showing log2 fold change (UPF3B_KO relative to Ctrl) and the y-axis showing -log10(FDR).

Methods - *Pathway Analysis (ORA)*

Data Sources (gprofiler2: Kolberg et al. 2020)

- **GO-BP** → biological processes
- **GO-CC** → cellular component, subcellular localization patterns
- **KEGG** → classical signaling & metabolic pathways
- **Reactome (REAC)** → pathways relevant to mRNA decay & neurodevelopment

Other Parameters

- **ordered_query = FALSE** → strict thresholded ORA (no gene ranking)
- **significant = TRUE** → return only statistically significant terms
- **exclude_iea = TRUE** → exclude automatic electronic annotations

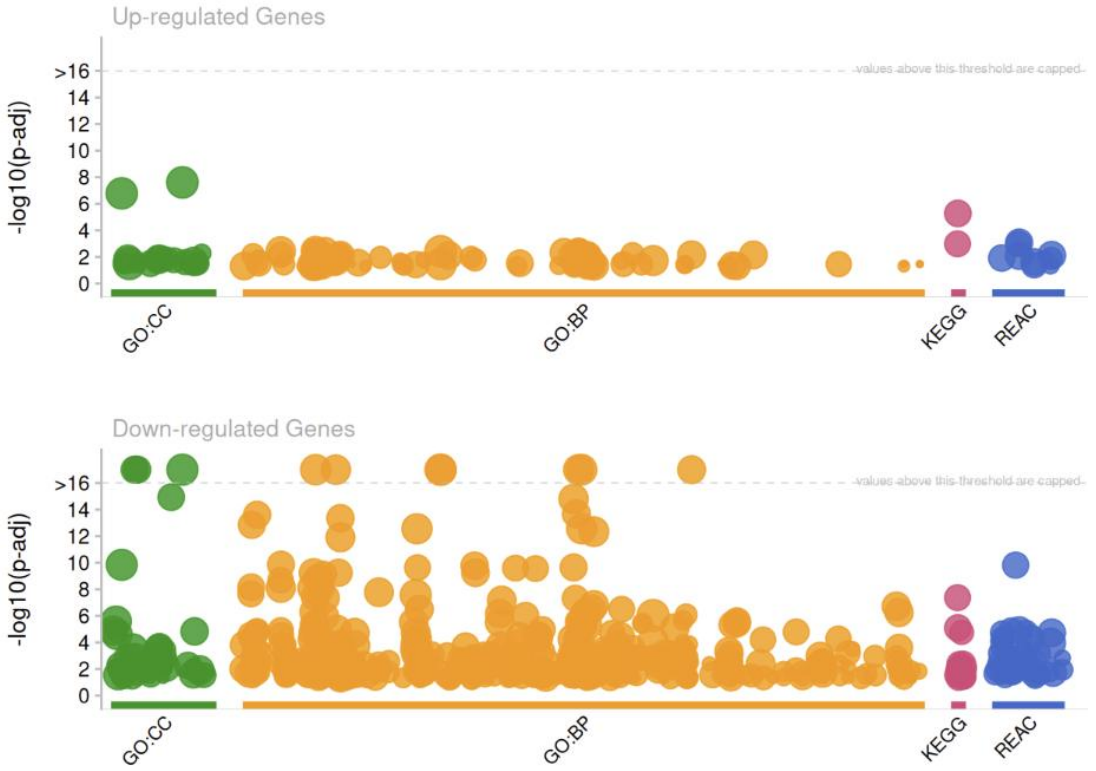
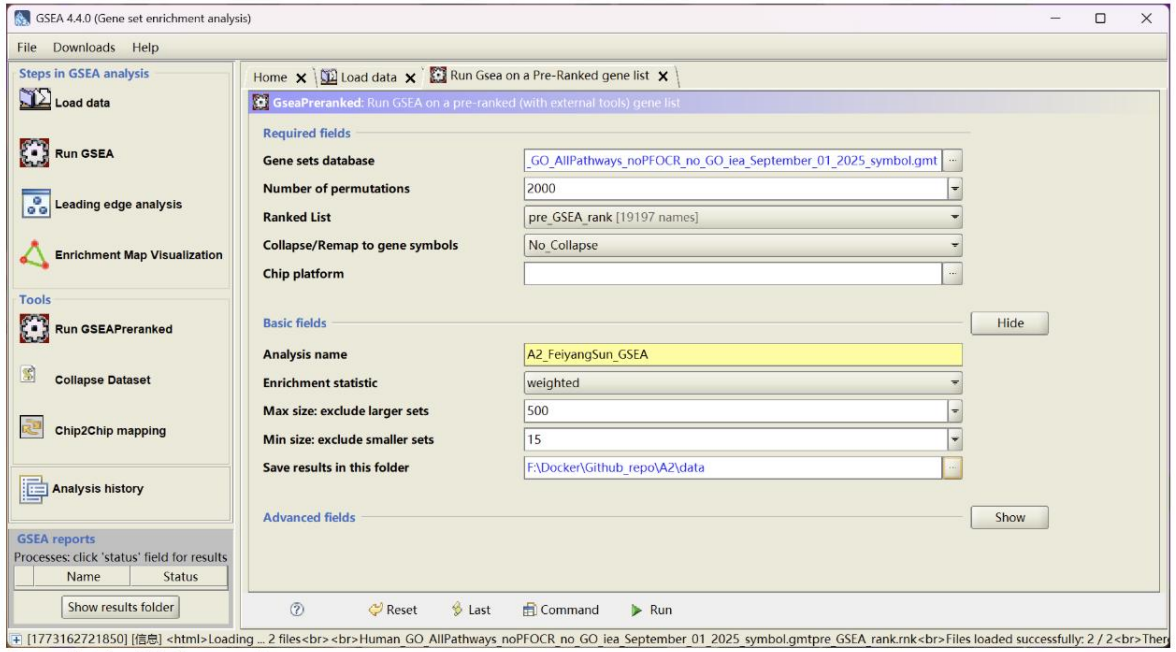


Figure 3. Manhattan plots showing enriched biological processes and pathways for up-regulated (top) and down-regulated (bottom) genes in human neural progenitor cells (NPC). The analyzed genes represent those potentially regulated by UPF3B, identified through differential expression analysis between UPF3B-KO and Ctrl samples.

Methods - *Pathway Analysis (GSEA)*

Parameters:

- GSEA desktop version: v4.4.0
- Geneset database: Human_GO_AllPathways_noPFOCR_no_GO_iea_September_01_2025_symbol.gmt (Bader Lab GeneSets database 2026)
- Number of permutations: 2000
- Ranked DEGs: pre_GSEA_rank.rnk
- **Ranking score = sign(log2FC) × -log10(FDR)**
- Collapse/Remap to gene symbols: No_Collapse
- Max size: 500
- Min size: 15



Scree shot for GSEA parameters

Figure 4. Screenshot of the parameter settings used for the GSEA analysis. The interface displays key configuration options including the number of permutations, selection of the ranked gene list file, and the option to disable gene symbol remapping since the ranked list already contains gene symbols.

Results - *GSEA (upregulated)*

	GS follow link to MSigDB	GS DETAILS	SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val	RANK AT MAX	LEADING EDGE
1	MITOCHONDRIAL TRANSLATION TERMINATION%REACTOME%R-HSA-5419276.5	Details ...	100	0.63	1.95	0.000	0.031	0.043	4043	tags=46%, list=21%, signal=58%
2	AEROBIC RESPIRATION%GOBP%GO:0009060	Details ...	158	0.58	1.91	0.000	0.044	0.117	3730	tags=32%, list=19%, signal=40%
3	ELECTRON TRANSFER ACTIVITY%GOMF%GO:0009055	Details ...	98	0.61	1.89	0.000	0.046	0.180	2027	tags=24%, list=11%, signal=27%
4	MITOCHONDRIAL TRANSLATION%REACTOME%R-HSA-5368287.5	Details ...	111	0.60	1.89	0.000	0.040	0.200	4043	tags=43%, list=21%, signal=54%
5	GLYCOSYLATION AND RELATED CONGENITAL DEFECTS%WIKIPATHWAYS_20250810%WP4521%HOMO SAPIENS	Details ...	22	0.80	1.88	0.000	0.033	0.209	2285	tags=50%, list=12%, signal=57%
6	REGULATION OF POSITIVE CHEMOTAXIS%GOBP%GO:0050926	Details ...	15	0.88	1.88	0.000	0.032	0.240	1350	tags=47%, list=7%, signal=50%
7	MITOCHONDRIAL INNER MEMBRANE%GOCC%GO:0005743	Details ...	414	0.53	1.88	0.000	0.028	0.248	4435	tags=39%, list=23%, signal=49%
8	OXIDOREDUCTION-DRIVEN ACTIVE TRANSMEMBRANE TRANSPORTER ACTIVITY%GOMF%GO:0015453	Details ...	59	0.65	1.87	0.000	0.032	0.308	1789	tags=24%, list=9%, signal=26%
9	MITOCHONDRIAL RESPIRATORY CHAIN COMPLEX ASSEMBLY%GOBP%GO:0033108	Details ...	89	0.61	1.86	0.000	0.031	0.331	4850	tags=52%, list=25%, signal=69%
10	CELLULAR RESPIRATION%GOBP%GO:0045333	Details ...	168	0.56	1.86	0.000	0.030	0.350	3732	tags=32%, list=19%, signal=39%
11	RESPIRATORY ELECTRON TRANSPORT%REACTOME DATABASE ID RELEASE 93%611105	Details ...	149	0.56	1.82	0.000	0.057	0.596	4638	tags=42%, list=24%, signal=54%
12	MITOCHONDRIAL RESPIRATORY CHAIN COMPLEX I ASSEMBLY%GOBP%GO:0032981	Details ...	38	0.69	1.82	0.000	0.053	0.605	4850	tags=63%, list=25%, signal=84%
13	CYTOCHROME COMPLEX%GOCC%GO:0070069	Details ...	29	0.72	1.81	0.001	0.062	0.689	3643	tags=52%, list=19%, signal=64%
14	RESPIRATORY CHAIN COMPLEX%GOCC%GO:0098803	Details ...	101	0.59	1.81	0.000	0.060	0.701	1897	tags=21%, list=10%, signal=23%
15	NADH DEHYDROGENASE COMPLEX ASSEMBLY%GOBP%GO:0010257	Details ...	38	0.69	1.81	0.000	0.059	0.718	4850	tags=63%, list=25%, signal=84%
16	OXIDATIVE PHOSPHORYLATION%GOBP%GO:0006119	Details ...	95	0.59	1.80	0.000	0.057	0.734	3533	tags=29%, list=18%, signal=36%
17	GLYCOLYSIS%PANTHER PATHWAY%P00024	Details ...	15	0.83	1.80	0.001	0.064	0.799	2017	tags=47%, list=11%, signal=52%
18	GLYCOLYTIC PROCESS%GOBP%GO:0006096	Details ...	30	0.70	1.79	0.000	0.065	0.826	2331	tags=33%, list=12%, signal=38%
19	ORGANELLE INNER MEMBRANE%GOCC%GO:0019866	Details ...	440	0.50	1.79	0.000	0.068	0.856	4435	tags=38%, list=23%, signal=49%
20	FATTY ACID CATABOLIC PROCESS%GOBP%GO:0009062	Details ...	74	0.60	1.79	0.000	0.065	0.859	3910	tags=38%, list=20%, signal=47%

Indirect support for neuron growth via mitochondrial metabolism.

Figure 5. Top 20 up-regulated pathways. These up-regulated gene sets (in UPF3B-KO) are primarily associated with mitochondrial function and energy metabolism, including pathways for mitochondrial translation, respiratory chain assembly, oxidative phosphorylation, and aerobic respiration.

Results - *GSEA (downregulated)*

extra-cellular matrix structure

Table: Gene sets enriched in phenotype na [plain text format]

	GS follow link to MSigDB	GS DETAILS	SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val	RANK AT MAX	LEADING EDGE
1	EXTRACELLULAR MATRIX STRUCTURAL CONSTITUENT%GOMF%GO:0005201	Details ...	97	-0.72	-2.16	0.000	0.000	0.000	1817	tags=46%, list=9%, signal=51%
2	EXTRACELLULAR MATRIX%GOCC%GO:0031012	Details ...	309	-0.62	-2.13	0.000	0.000	0.000	1729	tags=35%, list=9%, signal=38%
3	EXTERNAL ENCAPSULATING STRUCTURE%GOCC%GO:0030312	Details ...	324	-0.61	-2.10	0.000	0.000	0.000	1729	tags=33%, list=9%, signal=36%
4	HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION%MSIGDBHALLMARK%HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	Details ...	149	-0.66	-2.10	0.000	0.000	0.000	2326	tags=47%, list=12%, signal=53%
5	CADHERIN SIGNALING PATHWAY%PANTHER PATHWAY%P00012	Details ...	122	-0.67	-2.05	0.000	0.000	0.001	1856	tags=42%, list=10%, signal=46%
6	EXTRACELLULAR STRUCTURE ORGANIZATION%GOBP%GO:0043062	Details ...	148	-0.65	-2.05	0.000	0.000	0.001	1879	tags=37%, list=10%, signal=41%
7	EXTERNAL ENCAPSULATING STRUCTURE ORGANIZATION%GOBP%GO:0045229	Details ...	147	-0.65	-2.05	0.000	0.000	0.001	1879	tags=37%, list=10%, signal=41%
8	EXTRACELLULAR MATRIX ORGANIZATION%GOBP%GO:0030198	Details ...	147	-0.65	-2.04	0.000	0.000	0.001	1879	tags=37%, list=10%, signal=41%
9	COLLAGEN FIBRIL ORGANIZATION%GOBP%GO:0030199	Details ...	47	-0.76	-2.03	0.000	0.000	0.001	2029	tags=51%, list=11%, signal=57%
10	CANONICAL WNT SIGNALING PATHWAY%GOBP%GO:0060070	Details ...	59	-0.72	-2.02	0.000	0.000	0.002	2118	tags=49%, list=11%, signal=55%
11	BONE MORPHOGENESIS%GOBP%GO:0060349	Details ...	30	-0.81	-2.01	0.000	0.000	0.002	1515	tags=50%, list=8%, signal=54%
12	EXTRACELLULAR MATRIX ORGANIZATION%REACTOME%R-HSA-1474244.5	Details ...	251	-0.60	-2.00	0.000	0.000	0.004	2346	tags=39%, list=12%, signal=44%
13	GLYCOSAMINOGLYCAN BINDING%GOMF%GO:0005539	Details ...	97	-0.66	-1.98	0.000	0.001	0.009	2288	tags=38%, list=12%, signal=43%
14	CELLULAR RESPONSE TO NUTRIENT%GOBP%GO:0031670	Details ...	23	-0.84	-1.97	0.000	0.001	0.014	1008	tags=39%, list=5%, signal=41%
15	PID_INTEGRIN1_PATHWAY%MSIGDB_C2%PID_INTEGRIN1_PATHWAY	Details ...	46	-0.73	-1.96	0.000	0.001	0.014	2326	tags=61%, list=12%, signal=69%
16	SMAD BINDING%GOMF%GO:0046332	Details ...	60	-0.70	-1.96	0.000	0.002	0.019	3483	tags=65%, list=18%, signal=79%
17	REGULATION OF FIBROBLAST PROLIFERATION%GOBP%GO:0048145	Details ...	41	-0.75	-1.96	0.000	0.002	0.023	2331	tags=49%, list=12%, signal=55%
18	WNT SIGNALING PATHWAY%GOBP%GO:0016055	Details ...	95	-0.65	-1.95	0.000	0.002	0.027	3111	tags=46%, list=16%, signal=55%
19	SKELETAL SYSTEM DEVELOPMENT%GOBP%GO:0001501	Details ...	205	-0.59	-1.94	0.000	0.003	0.037	1817	tags=36%, list=9%, signal=39%
20	GDNF RET SIGNALING AXIS%WIKIPATHWAYS_20250810%WP4830%HOMO SAPIENS	Details ...	22	-0.83	-1.93	0.000	0.003	0.041	1468	tags=55%, list=8%, signal=59%

Wnt signaling (#10 & #18) – regulates neurodevelopment, neuronal differentiation, and axon guidance.

GDNF RET signaling (#20) – critical for dopaminergic neuron survival, differentiation, and axon growth.

Figure 6. Top 20 down-regulated pathways. The top 20 down-regulated gene sets (in UPF3B-KO) are primarily associated with extracellular matrix structure, organization, and cell adhesion, including collagen fibril organization, cadherin signaling, and integrin-mediated pathways.

Results - *Network visualization* *Cytoscape* (Shannon et al. 2003)

Creating the Enrichment Map

EnrichmentMap (Merico et al. 2010)

- FDR adjusted p-Value Cutoff: 0.1
- Edge Cutoff: 0.375
- 412 nodes and 1078 edges

Cluster creation

Perform clustering via *clusterMaker2* (Morris et al. 2011)

Generate node labels using *WordCloud* (Oesper et al. 2011)

Key Clusters:

Wnt & Morphogenesis – neural differentiation, axon guidance, synapse formation

ECM – neural structure and tissue organization

Coordinated down-regulation of neurodevelopmental and structural pathways in UPF3B knockout cells

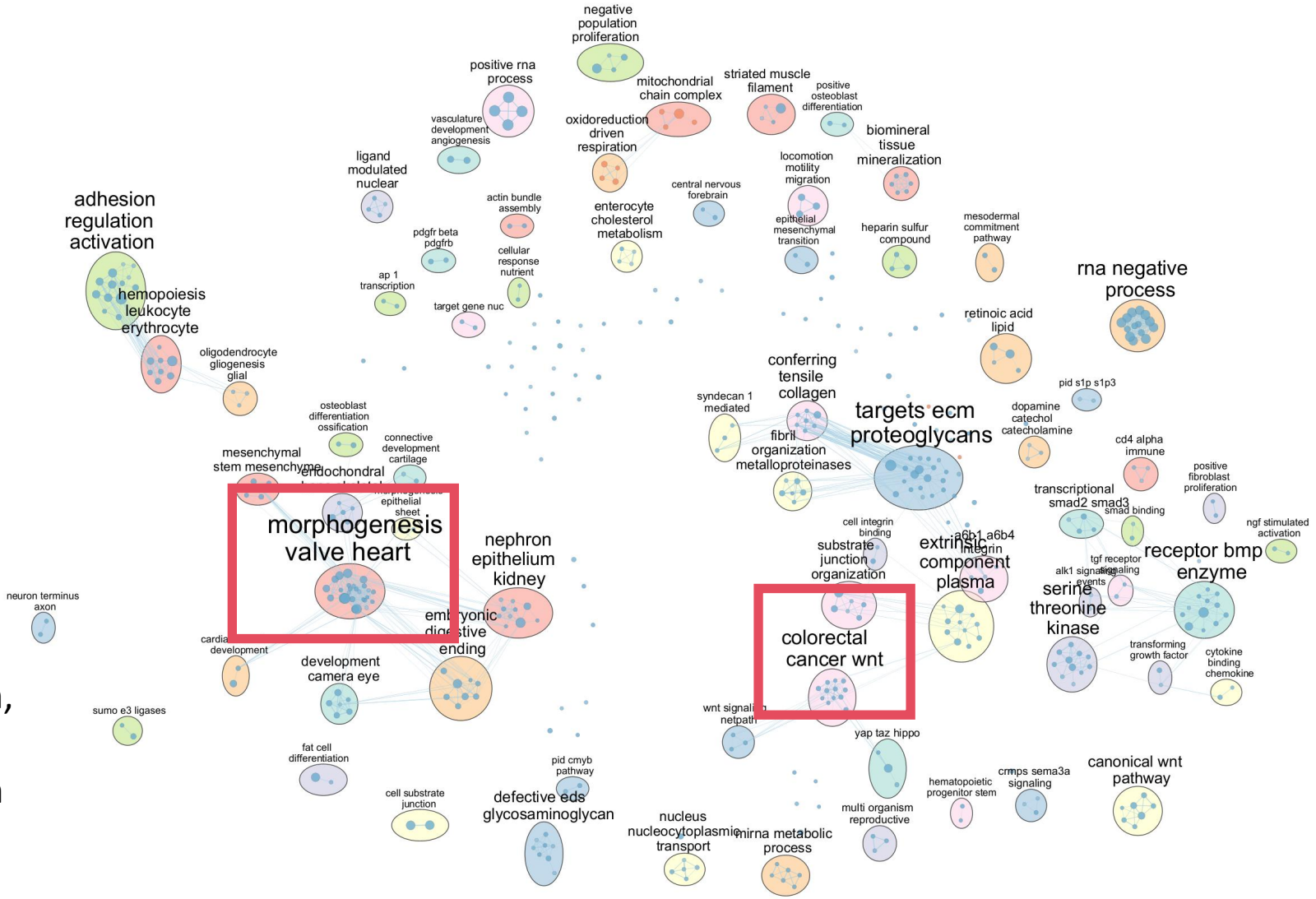


Figure 7. Functional clustering of enriched gene sets after layout. Application of a network layout improved the spatial separation and organization of clusters, facilitating clearer visualization of functional themes and relationships among enriched gene sets.

Discussion and Key insights

Pathway changes:

Most pathways down-regulated; mitochondrial/metabolic pathways up-regulated

Neurodevelopmental impact:

Wnt, morphogenesis, ECM → impaired neural differentiation, axon guidance, synapse formation

Network visualization:

Coordinated down-regulation of key clusters

Literature support:

GDNF/RET signaling – dopaminergic neuron survival and axon guidance (Chamieh et al. 2008)

Wnt signaling – neuron proliferation and axon outgrowth (Alrahbeni et al. 2015)

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